# Text properties and children's reading pace in a multilingual storytelling app

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# Abstract

Children increasingly read machine-generated stories on personal devices, and it is not known whether the text properties that govern reading under controlled conditions govern it there. In this Registered Report, we will analyse behavioural logs from *Let's Story*, a storytelling feature of the *Applaydu* app, in which children aged 7 and older read in ten or more languages. We will model reading time per word from each text's lexical difficulty, grammatical load and referential continuity, computed with language-general tools. Six confirmatory hypotheses concern these associations, their moderation by age band, the selectivity of read-aloud narration, children's own difficulty ratings and accumulated in-app reading. Bayesian crossed-effects models will represent readings within texts, children and languages, and every association will be judged against a region of practical equivalence. Because narration, text choice and continued use are chosen by families, we will report every coefficient as a conditional association.

# 1 Introduction

A generation of children now meets a large share of its reading on a screen, and an increasing share of what appears there was written by a machine. Commercial storytelling applications generate a narrative to order from a handful of choices a family makes, read it aloud in the family's language, and record what the child did with it. The behavioural record such platforms accumulate is orders of magnitude larger than any laboratory corpus of children's reading, spans dozens of languages, and arises under exactly the conditions that laboratory work has to strip away. It is also almost entirely unexamined. We ask whether the text properties that govern reading under controlled conditions still govern it in the wild, and whether their hold on the reader changes over the years in which decoding becomes automatic.

The laboratory answer is well established at three levels of linguistic description. At the level of the word, frequency of occurrence is among the most robust predictors of reading behaviour that psycholinguistics has produced. High-frequency words are recognised faster and fixated for less time than low-frequency words, across dozens of eye-tracking studies and with remarkable consistency across languages and orthographies[1,2]. In children the effect tends to be larger than in adults, reflecting less automatised lexical access[3]. Frequency norms drawn from subtitle corpora approximate everyday exposure well enough to have become the standard operationalisation[4,5]. Resources built from what children actually read, such as the Children and Young People's Books Lexicon, give a still more defensible baseline for developing readers[6]. Related lexical properties behave similarly. Concrete and highly imageable words are processed faster and remembered better than abstract ones[7], an advantage usually attributed to the availability of perceptual as well as verbal

representations[8,9]. The advantage is pronounced in younger readers, who lean more heavily on grounded representations when background knowledge runs out[10]. The diversity of the vocabulary a text deploys cuts differently according to who is reading: varied word choice may sustain a skilled adult, whereas for a child still spending resources on decoding it can overwhelm them[11–13].

At the level of the sentence, structure imposes its own cost. Mean dependency distance, the count of embedded clauses and the ratio of subordinate to main clauses all index the demand a text places on the reader to hold a dependent in mind until its head arrives[14]. Long dependencies are expensive because the intervening material has to be carried, and because children's verbal working memory is still developing they are disproportionately affected by that cost relative to adults[15–17]. Sensitivity to grammatical structure continues to underwrite comprehension well past the primary years: Marjokorpi and van Rijt[18] found explicit grammatical understanding to predict reading comprehension in secondary-school students after socioeconomic background, writing ability and other literacy factors were controlled.

At the level of the passage, what matters is whether the text states the connection between one idea and the next or leaves the reader to supply it. Sentence-to-sentence semantic similarity, overlap in referring expressions and the density of explicit connectives together index how much of that work the surface of the text has already done[19]. Texts that do more of it are generally easier to follow, and the benefit is largest for readers with least prior knowledge to draw on[20]. Automated tools now make these indices computable at scale[21,22], and multi-dimensional characterisations of difficulty built from them outperform the classical readability formulae they replaced[23–26].

The systematic review that accompanies this protocol sharpens that general account in ways that shape the present design. Text difficulty does not reduce to word frequency: in 924 German fourth-

graders, estimated lexical coverage predicted comprehension non-linearly, rising more steeply above roughly 56 per cent coverage[27]. In 158 children aged between 8 years 8 months and 11 years 2 months, word frequency, concreteness, deep cohesion, narrativity and lexical arousal each explained unique variance in passage comprehension after decoding and oral language were taken into account[28], which is the direct warrant for carrying affective and discourse variables alongside conventional readability measures. Effects also depend on what is being asked: across 1,229 children in grades 2 to 9, the direction and size of a text characteristic's contribution varied with question type[29]. Feature-based prediction of item-level comprehension from sentence complexity, lexical surprisal and rare-word use is already feasible[30], and syntactic complexity has been shown to impair comprehension accuracy and slow recall in bilingual children[31].

Two further strands bear on the present design because they concern the reader and the reading context. The first is the child's history of reading. Longitudinal work is unambiguous that this history cannot be read as a simple dose of practice. Using random-intercept cross-lagged models from ages 7 to 16,[32] found that weaker early reading predicted less subsequent leisure reading, while later within-person increases in book reading predicted gains in comprehension. A ten-year study likewise found reciprocal associations between skill and print exposure, with the skill-to-practice path most visible early and accumulated exposure becoming more predictive later[33]. Accumulated use of an app therefore carries both practice and the stable differences in skill and motivation that produced the use in the first place, so we will estimate within-child change and between-child differences separately.

The second is narration. Read-aloud audio is often treated as a uniform benefit, and the evidence does not support that reading. A meta-analysis of 30 reading-while-listening studies (62 effects,

total $N$ = 1,945) found a small pooled comprehension advantage ($g$ = 0.18), no reliable advantage where ordinary reading was self-paced ($g$ = 0.06, 95% CI [-0.07, 0.19]), and a larger one only where ordinary reading was paced by the experimenter ($g$ = 0.41, 95% CI [0.13, 0.70])[34]. Among students with reading difficulties, text-to-speech tools showed a moderate pooled effect ($d$ = 0.35) that fell to $d$ = 0.15 and included zero in within-participant designs[35]. Neither synthesis settles the mechanism, which a large naturalistic corpus is well placed to address. If a spoken rendition works by supplying word identity without print-to-sound mapping, it should remove much of the cost that lexical difficulty imposes while leaving the cost of connecting ideas largely intact. If narrated reading time is instead set by the duration of the recording, every text property should lose its purchase on time in the same proportion. Hypothesis 4 is written to separate these accounts.

The absence of such analyses can be counted. On 26 August 2026, a Scopus search over 2018 to 2026 returned 3,701 records on children's reading comprehension, 6,592 on computational text features and 133,132 on platform log data. The pairwise intersections were 17, 28 and 45 records. The three-way intersection was empty. The exact expressions behind those counts are reproduced in the supplementary search record, and Figure 1 shows the sets. Over the same window, records combining children's reading with machine-generated text went from none before 2024 to two, six and four in the three years since. The material children read is changing faster than the evidence base that describes it.

*Let's Story*, a feature of the *Applaydu* app developed by Gameloft and Ferrero International, sits in that empty intersection. Families choose a setting, a hero, a companion, a plot and a theme. A generative model then writes a narrative around those choices, and the app renders it as an illustrated storybook and, at the family's option, reads it aloud[36]. Every completed reading is logged, with the

time taken, the language, the child's age band, the reading level the family has set and an optional judgement of how hard the story was. Comparable platforms have been studied one feature at a time.[37] reviewed 11 studies and found read-aloud narration to aid comprehension while additional interactive games did not. Content analyses of the wider market are less encouraging:[38] found expert reviewers willing to recommend only 27.5 per cent of 309 phonics apps, and[39] found educational content the weakest dimension of 50 Google Play apps and consumer star ratings a poor guide to expert quality. Log data from a research-oriented platform have been shown to track ability, with reading speed in the BookBites app correlating with standardised scores at $r = .39$ to $.48$[40], but the commercial edutainment sector, whose user base is far larger, has not been examined this way at all.

This study is observational throughout. Children choose their stories, their reading level and whether to switch narration on, and families choose whether to keep using the app. Nothing here identifies a causal effect, and every coefficient is reported as an association conditional on the adjustments made. Three consequences of that stance shape the design itself. First, the reading level a family sets determines which version of a text the platform serves, so text difficulty is partly assigned by a choice that also reflects the child's capacity. We will therefore estimate the text coefficients within reading level, and a pre-specified check asks whether they survive that restriction. Second, the app records elapsed time for a whole story and never for a page, so the outcome is a rate, not a fixation record, and we state below which analyses that forecloses. Third, the data carry no measure of any individual child's reading skill, vocabulary or language background, so age band stands in for a bundle of developing capacities and is interpreted as nothing finer.

The theoretical commitments behind the hypotheses are held in a machine-checkable theory object, deposited with this protocol, that names twelve constructs, eighteen propositions each carrying the

mechanism it claims, six auxiliary assumptions and six rival accounts, and records which prediction is derived from which proposition and which rival it is meant to separate. Figure 2 is generated from that object, so the figure and the hypotheses cannot drift apart. The object passes all twelve items of a versioned rigour checklist, scoring 90.9 out of 100 (Supplementary Section S1)[41]. Its causal graph is acyclic, so it yields testable implications: that the three text properties are mutually independent given how a text came to be written, that narration is unrelated to accumulated reading, and that the difficulty rating adds nothing about reading level once reading time is known. We will test these implications before interpreting the confirmatory models, and they can fail.

Six confirmatory hypotheses follow, set out with their models, decision rules and interpretations in Table 1. Two of them, the interactions between the text properties and the age band, are registered as two-sided tests that adjudicate between rival accounts, because the direct developmental evidence in this age range supports neither sign. Each account, and the reading of every possible outcome, is given where the hypotheses are stated.

# 2 Hypotheses

Three text components carry the confirmatory predictions. **LexicalDifficulty** and **Grammatical-Load** are oriented so that higher scores mean a more demanding text, and **ReferentialContinuity** so that higher scores mean a text that states more of the links between its ideas. Signs are fixed before any component score is joined to an outcome. The age band is coded 0 for children recorded as 7 to 8 and 1 for children recorded as 9 or older, and narration is coded 0 when the spoken rendition was off and 1 when it was on. The outcome for Hypotheses 1, 2, 3 and 6 is log reading time per word, standardised within language, so a coefficient is a within-language standardised slope and the region of practical equivalence is 0.05 standard deviations. Hypothesis 4 concerns a difference between two attenuation proportions, with a region of 0.05 on that difference. Hypothesis 5 uses the child’s own three-point difficulty rating on an ordinal scale, where the region is an odds ratio between 0.95 and 1.05.

Because the two main effects run in opposite directions, the same developmental claim takes opposite signs in the two interaction terms. Grammatical load slows reading, so a weaker effect in older children makes a positive slope smaller and the interaction negative. Referential continuity speeds reading, so a weaker effect in older children makes a negative slope less negative and the interaction positive. Wherever an account is described below as predicting that a text property bears less on the older reader, the sign of the corresponding coefficient follows from this convention and not from a separate claim.

Two features of the outcome do work that the hypotheses depend on, and both are fixed here. First, children in the two age bands differ in overall reading rate, and a difference in rate that is purely

multiplicative manufactures a spurious interaction on a raw time scale[42]. The outcome is therefore the logarithm of reading time throughout, on which a multiplicative difference in rate appears as a main effect and never as an interaction, so that every interaction reported here is a difference in *proportional* cost. Second, in the standard cohesion manipulation the more cohesive version of a text is 20 to 40 per cent longer, so a duration aggregated over a whole text would show higher coherence slowing reading for arithmetic reasons alone. Dividing by word count and retaining log text length as a covariate closes that path, and a raw duration is never used as an outcome.

A hypothesis is declared untestable, before its model is fitted and without reference to any coefficient, if its component fails the construction rule set out under the dimension reduction or if its outcome or contrast is absent from the frozen extract. A hypothesis whose data fall below the precision threshold recorded for it in Table 1 is instead declared underpowered: its model is still fitted and reported, and no conclusion is drawn from an interval the design analysis predicted would be indeterminate.

**Hypothesis 1. Text properties predict reading pace.** We predict that, across distinct texts and within reading level and language, LexicalDifficulty and GrammaticalLoad will each show a positive association with log reading time per word, and ReferentialContinuity a negative one. The three coefficients are evaluated and labelled separately. An overall verdict on Hypothesis 1 is recorded only when all three are in the predicted direction and practically non-negligible, and the failure of one does not suppress reporting of the others.

**Hypothesis 2. Grammatical load and the age band: capacity against experience.** We register the GrammaticalLoad × AgeBand interaction as a two-sided adjudicating test, because two accounts of what changes between these ages predict opposite signs on the scale this study uses.

On the capacity account, a grammatical dependency costs time in proportion to how close its

maintenance comes to exhausting the working memory available, so as that capacity matures the same structure costs proportionally less and the interaction is negative[15,43]. On the experience account, what changes is not capacity but exposure: an older reader has met complex structure far more often and has been taught the punctuation that marks its boundaries, engages with it more fully in consequence, and pays a proportionally larger cost, which makes the interaction positive[44,45].

The direct evidence is open. The best-powered test over this exact band found no attenuation at all, but it manipulated temporary ambiguity, which indexes reanalysis, not the sustained maintenance that dependency length imposes, so its null does not transfer to the present operationalisation[46]. Where an interaction has been reported it was confined to late measures with fourteen children per group, and in that same study the complexity effect expressed as a proportion of baseline *grew* with age, from 15 per cent in the younger children to 25 per cent in the older and 31 per cent in adults[47]. Since the outcome here is log reading time per word, its coefficients are proportional, and that growth is exactly the pattern a positive interaction would reproduce. A third possibility is that development over this band is not monotonic at all[48].

A negative interaction outside the region of practical equivalence supports the capacity account, a positive one supports the experience account, and a posterior lying inside the region is evidence that neither developmental signature reaches a detectable size across these two bands. That third outcome would still be informative: no existing test had more than 81 children.

**Hypothesis 3. Referential continuity and the age band: knowledge against cue use.** We register the ReferentialContinuity × AgeBand interaction the same way, because the construction-integration framework supplies two moderators of coherence facilitation that both rise between 7 and 11 and push in opposite directions.

On the knowledge account, a reader who cannot bridge a gap from what they already know depends on the text to state the connection, so surface continuity is worth most to the younger reader and the interaction is positive, the facilitation being a negative slope that becomes less negative with age. On the cue-utilisation account, exploiting a connective requires enough decoding fluency to reach it and enough explicit knowledge of what it signals, so the older reader gains more and the interaction is negative[49].

Here too the existing evidence does not decide between the accounts. Eight- and ten-year-olds gained the same speed-up from an appropriate connective in self-paced reading, with the age interaction explicitly null[50]. Online sensitivity to a coherence break did not differ between children of 8 to 9 and children of 10 to 11[51]. The single demonstration of a reverse cohesion effect in children turned on world knowledge within a single grade, and neither age nor decoding skill played any part in it. No study has manipulated whole-text coherence in this age range and measured reading rate, so that cell of the literature is empty.

The three outcomes are read as for Hypothesis 2. Because the sign of a coherence effect also depends on which cohesive device carries it, connectives enter by relation type and the component's anchors are the causal and contrastive connectives, for the reason given under the dimension reduction.

**Hypothesis 4. Narration removes lexical cost selectively.** We predict that narration attenuates the LexicalDifficulty coefficient proportionally more than the ReferentialContinuity coefficient. Writing $\pi_{\text{lex}}$ and $\pi_{\text{cont}}$ for the proportional attenuation of each coefficient under narration, computed draw by draw from the posterior, the prediction is that $\pi_{\text{lex}} - \pi_{\text{cont}} > 0.20$. A spoken rendition supplies word identity without print-to-sound mapping, which is the step that a low-frequency or long word makes expensive, and it does not supply the link between two ideas, which the reader

must still construct. Reading aloud only the decoding-heavy elements of a test helps poor decoders while a pacing-only control does nothing[52]. The benefit is confined to children whose decoding is weak but whose language is intact[53], and within a narrated passage the time saved falls on the first encounter with an orthographically novel word and has gone by the second[54].

The competing account holds that narrated reading time is set by the duration of the recording, in which case every text coefficient shrinks by the same proportion and $\pi_{\text{lex}} - \pi_{\text{cont}} = 0$. That account has substantial support of its own. The aggregate comprehension benefit of reading while listening is confined to studies whose comparison condition was externally paced[34,55]. A preregistered test of the resource-freeing mechanism found comprehension *worse* under narration, and attributed it to a forced pace blocking rereading[56]. Where the compression has been measured incidentally it is severe: the standard deviation of exposure time fell from 11.56 seconds when self-paced to 0.62 seconds when narrated[54]. No study has yet regressed reading time on decoding-level and on integration-level text predictors separately, in narrated and unnarrated reading of the same texts, so neither account has been tested on the contrast that would separate them.

Two safeguards follow. First, the model will estimate a separate residual standard deviation for narrated and for unnarrated sessions, and we will report that ratio alongside every Hypothesis 4 estimate. A large fall in residual variability under narration is the signature of a recording setting the pace, and we will not interpret the selectivity contrast without it. Second, we will repeat the contrast in the subset of narrated sessions in which the audio cannot have been binding, meaning those read faster than 160 words per minute, a rate below the slowest adult silent reading rate recorded across languages and therefore a conservative ceiling on any narration rate. If the selectivity survives where the recording was not the constraint, pacing does not explain it. Should the data controller

supply the audio duration, that bound will be replaced by the duration itself and we will report the substitution as a deviation from this protocol.

Because the contrast is a ratio, we will evaluate it only when the lexical and continuity coefficients from Hypothesis 1 are both practically non-negligible in sessions without narration. If either is not, there is no attenuation to compare and Hypothesis 4 is reported as untestable. This gating rule does not depend on the sign of any coefficient.

**Hypothesis 5. Text properties predict the child's own judgement of difficulty.** We predict that, in an ordinal model of the post-reading rating, LexicalDifficulty and GrammaticalLoad will raise the odds of a harder rating and ReferentialContinuity will lower them, each by an odds ratio outside 0.95 to 1.05, and that the three standardised coefficients will rank by magnitude as they do in the reading-time model. The second clause makes this more than Hypothesis 1 repeated on a second outcome. Two measures of the same three constructs, one behavioural and one introspective, obtained on different scales from overlapping sets of children, should rank those constructs the same way, and we will report the posterior probability that they do as the study's principal convergent-validity check.

Two facts about the rating constrain the design, and neither is established in the literature. No published study has had children of this age give a global post-reading judgement of how difficult a text was and validated it against the text's measured properties: the child metacognition literature rates adjacent things, such as confidence in one's own answers, judgements of learning for word lists and enjoyment of an activity. The three-point easy, just right and difficult format is a classroom heuristic with no published psychometric validation, and nothing licenses assuming that a single rating from a single child is reliable. That children can validly judge text difficulty is therefore

something this study tests, not something it assumes.

For the design, the effective sample size is the number of distinct children rating each text, and the raw count of ratings overstates it, so the model carries a text varying intercept and we will report the distribution of raters per text before fitting. A text rated by fewer than five distinct children does not contribute. We will report the intraclass correlation for the rating at text level, the reliability estimate the literature does not supply.

For interpretation, a failure here is ambiguous on its own, and Hypothesis 1 resolves it. If the text components predict reading time but not the rating, the feature pipeline is sound and the rating is not a usable measure of experienced difficulty, which is worth reporting in its own right given how widely such single-item checks are deployed in educational software. If they predict neither, the components themselves are suspect. If they predict both and in the same order, two independent measurements converge on the same ordering of text demands. All three readings are recorded here in advance.

**Hypothesis 6. Accumulated reading is followed by a faster pace.** We predict that, among children with several recorded sessions, a doubling of a child's own accumulated in-app reading will be followed by a reduction in that child's log reading time per word, over and above session order and the interval since the previous session. Every encounter with a written word form sharpens the representation supporting its later recognition, so reading consolidates the machinery that makes reading cheap. The relation between practice and skill is reciprocal over years[32,33], so the coefficient is an association within the observation window and not an estimate of a practice effect. This hypothesis depends on a subsample whose size is not known before the extract is released, and it is declared underpowered, not untestable, if that subsample falls below the threshold in Table 1.

Every hypothesis is labelled under the single posterior rule given under Decision rules and diagnostics, and an inconclusive result is reported as such. No hypothesis is filtered by a preliminary significance test. The one registered exception is the Hypothesis 4 gating rule above, under which that hypothesis is reported as untestable.

Table 1 sets out each hypothesis against the sampling and analysis plan that tests it, the precision the design analysis says that test requires, how each possible outcome will be read, and what would be shown wrong. Every row was fixed before any outcome data existed.

**Table 1**
*Study design table*

| Question | Hypothesis | Sampling plan | Analysis plan | Sensitivity of the test | Interpretation given different outcomes | Theory that could be shown wrong |
|---|---|---|---|---|---|---|
| Do the text properties that govern reading in the laboratory govern it in an app? | H1. LexicalDifficulty and GrammaticalLoad are positively, and ReferentialContinuity negatively, associated with log reading time per word. | Tier 1: every eligible completed reading in the frozen extract, one row per child and text, children recorded in age band 7-8 or 9+. No stopping rule, and the extract is fixed before modelling. | The tier-1 Bayesian model specified under Confirmatory models: within-language standardised log time per word on the three components, the age band, narration and their interactions, adjusted for reading level, text length, valence, arousal, narrativity and provenance, with varying intercepts for Text, Language, Country and App version, varying component slopes by Language, and an Account intercept where the estimability rule is met. The three component posteriors are the registered tests. | ROPE of 0.05 *SD*. Simulation over the number of distinct texts puts a determinate verdict on an effect of 0.10 *SD* at 738 distinct texts for the lexical coefficient (95 per cent interval 640 to 843) and fewer for the other two, with probability 0.90. Below that the hypothesis is declared underpowered before fitting. | Each coefficient is labelled separately as practically non-negligible, practically equivalent to zero, or inconclusive. A combined H1 verdict is recorded only when all three are in the predicted direction and non-negligible. | That the lexical, grammatical and discourse determinants of reading established under laboratory control operate at all on unsupervised reading of machine-generated stories. |

| Question | Hypothesis | Sampling plan | Analysis plan | Sensitivity of the test | Interpretation given different outcomes | Theory that could be shown wrong |
|---|---|---|---|---|---|---|
| Does grammatical demand weigh proportionally less, or more, on older children? | H2. GrammaticalLoad × AgeBand, tested two-sided. The capacity account predicts a negative interaction, the experience account a positive one. | As H1. Tier 1 admits only readings with a recorded age band, so both bands are represented by construction. | The same tier-1 model. The GrammaticalLoad × AgeBand posterior is the registered test. | ROPE of 0.05 *SD*. A determinate verdict on an interaction of 0.10 *SD* needs 1,230 distinct texts (1,107 to 1,359) at twelve readings each. Declaring practical equivalence when the interaction is truly zero needs about 1,600. Cross-validated comparison against a model without the interaction is reported alongside and does not gate the coefficient. | Negative and outside the ROPE supports the capacity account, and positive and outside supports the experience account. Inside the ROPE is evidence that neither developmental signature reaches a detectable size across these bands, which no existing test of at most 81 children could have established. | Whichever account the sign contradicts. A posterior inside the ROPE weighs against both, since each predicts a signed proportional change. |

| Question | Hypothesis | Sampling plan | Analysis plan | Sensitivity of the test | Interpretation given different outcomes | Theory that could be shown wrong |
|---|---|---|---|---|---|---|
| Does explicit connection between ideas help the younger or the older reader more? | H3. ReferentialContinuity × AgeBand, tested two-sided. The knowledge account predicts a positive interaction, the cue-utilisation account a negative one. | As H2. | The same tier-1 model. The ReferentialContinuity × AgeBand posterior is the registered test. Connectives enter the component by relation type, with causal and contrastive connectives as anchors. | ROPE of 0.05 *SD*. A determinate verdict needs 1,431 distinct texts (1,276 to 1,596), and declaring practical equivalence needs about 1,600. | Positive and outside the ROPE supports the knowledge account, on which continuity substitutes for an inference the younger reader finds costly. Negative and outside supports the cue-utilisation account, on which a connective must first be reached and understood before it can help. Inside the ROPE is evidence for neither. | Whichever account the sign contradicts. No study has yet manipulated whole-text coherence in this age band and measured reading rate. |
| How does read-aloud narration work? | H4. The proportional attenuation of the lexical coefficient under narration exceeds that of the continuity coefficient by more than 0.20. | As H1. Evaluated only where both narration states occur within the eligible sample and, in the sensitivity analysis, within the same child. | Draw-by-draw contrast of the two proportional attenuations, computed from the narration interactions of the same tier-1 model. Gated on the lexical and continuity coefficients both being practically non-negligible without narration. | ROPE of 0.05 on the difference of proportions. Narration is heavily unbalanced in these data, so the required number of texts is the simulated figure multiplied by the design effect $1/(4p(1-p))$ at the observed prevalence $p$, reported before fitting. | A difference above 0.20 supports selective removal of decoding cost. A difference practically equivalent to zero supports the account on which the recording sets the pace. Anything between is inconclusive and both accounts remain live. | Either the decoding-scaffolding account of narration or the audio-pacing account. The contrast is constructed so that they cannot both survive. |

| Question | Hypothesis | Sampling plan | Analysis plan | Sensitivity of the test | Interpretation given different outcomes | Theory that could be shown wrong |
|---|---|---|---|---|---|---|
| Do children's own difficulty judgements track the same text properties? | H5. The three components predict a harder rating with odds ratios outside 0.95 to 1.05, and rank in the same order as in the reading-time model. | Tier 1 sessions carrying a rating. The rating is optional, so missingness is high and is modelled explicitly, never assumed ignorable. The effective sample size is distinct children per text, and a text rated by fewer than five does not contribute. | Ordinal cumulative-logit model with the same fixed and grouping structure. The ordering claim is the posterior probability that the three coefficients rank by magnitude as they do for reading time. | ROPE of an odds ratio between 0.95 and 1.05, a half-width of 0.049 on the log-odds scale. Simulated separately with an ordinal mixed model, at the 30 per cent rating availability the structural sample showed and with the five-rater floor applied. The most demanding requirement in the design: the 0.90 target is not reached within the 1,600 texts simulated, and on the established inverse-square-root scaling it projects to roughly 8,000 to 8,700 generated texts, about 2,300 to 2,500 carrying at least five ratings. The simulation is re-run at the observed availability rate from the provider counts before fitting, and this hypothesis is the one most likely to be declared underpowered. Because ratings are missing for most sessions, an inverse-probability | Agreement on sign and order means two independent measurements converge on the same ordering of text demands. Disagreement, with the reading-time coefficients themselves non-negligible, indicates that the rating is not a usable measure of experienced difficulty, not that the components are wrong. Failure on both outcomes indicts the components. | That reading time and reported difficulty are both readouts of the same expended effort. |

| Question | Hypothesis | Sampling plan | Analysis plan | Sensitivity of the test | Interpretation given different outcomes | Theory that could be shown wrong |
|---|---|---|---|---|---|---|
| Does reading more in the app go with reading faster? | H6. Within a child, accumulated in-app reading is negatively associated with log time per word. | Tier 2: children with at least three eligible sessions on distinct texts. Declared underpowered, not untestable, if tier 2 holds fewer than 563 children (95 per cent interval 463 to 674). | The tier-1 model refitted on tier 2 with a child varying intercept, adding the within-child deviation in accumulated words and the child's mean, plus session order and days since the previous session. The within-child term is the registered test. | ROPE of 0.05 *SD*. Simulation over the number of children in tier 2 at five sessions each. | A negative within-child coefficient is reported as an association over the observation window, never as a practice effect, because skill and practice are reciprocal over longer spans. | That reading consolidates the recognition machinery reading depends on, within a window short enough for the reverse path to be unlikely to dominate. |

# 3 Methods

Two conventions hold throughout this section. First, every decision rule in the protocol is a function of provider-computed counts, of written answers from the data controller, or of thresholds registered here, and never of a fitted coefficient. Second, wherever the text says a quantity is reported, we will report it in the Stage 2 manuscript whether or not it changes a conclusion.

## 3.1 Ethics

Data are generated in the ordinary operation of a commercial application and are collected by its publisher under the terms of use and privacy policy a family accepts on installation, with consent for research use of anonymised records registered in the app by a parent or guardian. We play no part in collection, have no contact of any kind with any child or family, and return nothing to the platform, so nothing done here can alter what any child experiences. The extract supplied to us carries no direct identifier: the account key is a pseudonymous string generated by the platform, and no name, contact detail, precise location or device identifier is included. Ethical approval for secondary analysis of these anonymised records was granted by the Departmental Research Ethics Committee of the Department of Education, University of Oxford (reference [to be inserted]). The study involves no experimental manipulation and no intervention, so it raises no question of withdrawal from a procedure. A family's right to have its records erased is exercised through the platform, and any erasure occurring between the building of the extract and its transfer will be recorded by the data controller and reported at Stage 2.

## 3.2 Data provenance, sequestration and outcome blindness

The data this protocol analyses already exist. Nature Human Behaviour admits secondary analyses of existing data on condition that authors can evidence no prior access to the data in question, and the arrangement set out below was put to the journal's editorial office and confirmed as appropriate before this protocol was written.

The data controller builds the export now, on its own release cycle, and holds it. Nothing of the session-level file reaches us until the journal confirms in-principle acceptance in writing. In the interval we receive three things, each of which the editorial office agreed may be received in advance. The first is a manifest: the column list, the row count, the cut-off timestamp, the build date and a SHA-256 checksum for every file, so that the file eventually transferred can be shown to be the file that was built. The second is the story texts, keyed by story identifier, reading language, reading level and app version, with page breaks preserved, because every linguistic measure in this protocol is computed on them and none of them contains a behavioural outcome. The third is a set of aggregate counts computed by the data controller from the export by running a script we supplied, which reads ten columns, writes one small table of counts, and never reports a reading time as a value.

The counts deserve detail, because a count computed from an outcome column is where blindness would fail. The script treats reading time only as a column whose usability has to be established. It reports how many values are absent, non-numeric, negative or zero, and how many fall into each of seven bands whose boundaries are successive powers of ten, together with the number of distinct values the column holds and the largest number of sessions sharing any single value. It never crosses

reading time with language, age, text, account or date. Bands ten times wider than one another cannot distinguish a fast text from a slow one. What they can distinguish is a column recorded in seconds from one recorded in milliseconds, which the analysis has to know before it can be specified at all. The count of sessions sharing one value is how a session-timeout cap makes itself visible, and the sampling plan below turns that count into a registered decision about censoring.

Two disclosures follow. First, between 27 July and 6 August 2026 we received a 35-row structural sample and the data dictionary, shared to confirm the shape of the table, together with the summary described above computed on that sample. The sample carries real reading times for 35 sessions on five distinct texts, one session per account. It cannot support any hypothesis in this protocol: none is identified by 35 observations, and every within-child term requires accounts with more than one session, of which it contains none. It is nonetheless part of the record, and we report it here. Second, at the editorial office's specific direction, we have not run the dimension reduction specified below and will not run it before submission, although we hold the texts on which it would operate. The component solution is therefore fully specified here and entirely unfitted.

The data controller will supply a signed certification naming an owner and a deputy for the held file, and stating its build date, its window, its row count and its checksums. The same document certifies three things about what has happened to the file: that it has not been transferred in whole or in part to the research team or to anyone acting for it, that it will be transferred only on written confirmation of in-principle acceptance, and that no analysis relating story text to reading time has been conducted by the controller or reported to the team. It is reproduced in the supplementary material.

## 3.3 Design and sampling plan

The design is observational and the sample is a census: we will analyse every eligible record in one immutable extract, whose closing date is fixed before any outcome model is fitted, with no sequential testing and no optional stopping.

Two nested samples serve different hypotheses, because those hypotheses are identified at different levels and the data supporting them are unequally certain.

Tier 1 carries the text hypotheses. It comprises every completed reading in the extract for which the account is recorded in age band 7-8 or 9 and over, the reading language is one the text-feature pipeline covers, the story text has been supplied, and the reading time falls inside the plausibility bounds below. Where an account read the same text more than once, only the first reading enters tier 1, and repeated readings are retained for the practice analysis described under sensitivity. The unit of analysis is one reading of one text by one account.

Whether tier 1 can estimate a child varying intercept depends on how many accounts read more than one distinct text, which is not known in advance. If at least ten per cent of tier-1 readings come from accounts contributing two or more readings, the model includes a varying intercept for Account. If fewer do, that term is not separable from residual variance: it is dropped, child variance is carried in the residual, and the reduction in the effective independence of the sample is stated wherever a tier-1 interval is reported. We will read the proportion from the provider-computed counts before fitting anything.

Tier 2 carries the within-child hypothesis. It comprises those tier-1 accounts contributing at least three eligible readings of distinct texts. Three is the smallest number that leaves a degree of freedom

once a within-child slope and an intercept are estimated, and the threshold is varied to two and to five in the sensitivity analyses. Tier 2 estimates a child varying intercept and separates each child's mean level of a predictor from that reading's deviation from it, so a within-child association is not conflated with stable differences between children who habitually read at different rates. How many accounts reach the tier-2 threshold is not known in advance. The structural sample contains none, but it holds one session per account by construction and cannot speak to the full window.

Reading time enters as elapsed seconds for the whole story divided by the story's word count, and we bound it by what a child could physically be doing, not by anything observed in these data. We will exclude a reading faster than 400 words per minute, or 0.15 seconds per word, as faster than reading. We will exclude a reading slower than 10 words per minute, or six seconds per word, as a story left open. The upper bound follows the only published precedent that states such a rule and its yield: a meta-analysis of Spanish reading rates deleted silent rates above 400 words per minute as exceeding what a competent adult can manage and as indicating skipped text, removing 313 of 19,237 cases[57]. The lower bound sits far below the slowest group mean in that literature, 30 words per minute for Spanish first-graders reading silently, so it cannot exclude a merely slow reader. Both bounds are wide by design, so the primary analysis removes only the impossible. Tighter treatments are sensitivity analyses. We will report the number and proportion removed by every rule, by language, age band, reading level and narration state.

One further rule concerns the top of the distribution. Where a session timeout caps the recorded interval, the cap shows itself as an implausible number of sessions sharing one value, which is one of the quantities the provider-computed counts report. If those counts show more than one per cent of sessions sharing a single value, we will locate that value in the transferred extract before

fitting any outcome model. If it lies in the upper decile of the reading-time distribution, we will treat it as a right-censoring point and fit the timing models as censored lognormal models, with those observations censored, not discarded.

Ahead of all of the above, we will exclude sessions the platform flags as incomplete or crashed, duplicated event records, accounts flagged as duplicates under the platform's own data-quality rules, and stories without a positive word count. A participant-flow diagram will report every exclusion and the retained accounts, readings and texts, overall and by language.

## 3.4 Materials

The reading materials are the story texts themselves. A family chooses a setting, a hero, a companion, a plot and a theme, and a generative model writes a narrative around that combination. The text is then fixed, and it is not rewritten when the same combination is chosen again. Texts also exist in a fixed classic set that families may read instead of a generated story. Alongside the texts, the extract records which of the two a reading drew on, the reading level the platform was serving, the language, the app version and the page count.

For identification, a text is a stimulus shared by many readers, not a property of a session, and every text coefficient in this protocol is identified by comparing texts. How many distinct texts the extract holds is therefore the single quantity that governs how much this design can establish, and it is not yet known: the data controller's own estimate spans a few hundred to a few million, according to whether one combination of chosen attributes maps to one text shared by everyone who picks it or to a separate text per session. The design analysis below is built around that uncertainty, and Table 2 records what follows from each answer.

For every text, we will compute features in four families. *Lexical* features are mean log frequency of content words, the proportion of words with a frequency below one per million tokens on the uniform frequency resource, mean rated age of acquisition, mean orthographic length, mean per-token surprisal under a language-specific neural language model, lexical density, the Measure of Textual Lexical Diversity and concreteness. *Grammatical* features are mean dependency distance from a universal-dependencies parse, clauses per sentence, the ratio of subordinate to main clauses, mean sentence length and mean depth of noun-phrase modification. *Discourse* features are mean semantic similarity between adjacent sentences from a multilingual sentence encoder, the proportion of adjacent sentence pairs sharing a referring expression, connectives per hundred words by relation type and narrativity. *Affective* features are the mean and the variability of valence and of arousal.

Supplementary Table 3 names the resource used for each feature in each language, with its coverage. Four rules govern that table and are fixed before any feature is computed. The first addresses a problem specific to multilingual work and unsolved in the literature: a variable measured by a different instrument in each language confounds the method of measurement with the language, so that a language-level slope no longer means the same thing twice over. Each feature is therefore computed from a single resource applied identically in every language, even where a better native resource exists in one of them, and native norms are run as a pre-specified sensitivity analysis with the per-language correlation between the two reported. Second, no value is ever carried across languages by translating a word list, and a predicted resource is admitted only where the prediction was made within each language by one documented procedure and its correlation with native ratings is reported. Where no acceptable measure exists for a language, that language does not contribute that feature. Third, every measure reports the proportion of tokens it matched, and a language in

which a measure matches fewer than 80 per cent of tokens does not contribute it. A separate per-text rule under the dimension reduction governs which texts enter the component solution. Fourth, the full inventory, the software versions, the model checkpoints, the tokenisation rules and the matched-token proportions will be frozen and published before any feature is joined to an outcome. Korean is the language these rules constrain most, and the consequences for it are set out beside that table.

## 3.5 Measures

The processing outcome, serving Hypotheses 1 to 4 and 6, is the natural logarithm of elapsed reading time divided by the story's word count, standardised within reading language. We describe it throughout as reading pace or processing time, never as a measure of comprehension or fluency, because nothing in these data licenses that reading. The outcome for Hypothesis 5 is the child's optional post-reading rating of the story as easy, just right or difficult, treated as ordered and modelled on a cumulative-logit scale.

The confirmatory predictors are the three component scores defined below. Text length in words, narrativity, mean valence and mean arousal are pre-specified secondary text predictors and enter every model as covariates. The variability of valence and of arousal, and the additive and temporal connective densities, are computed for description and enter no confirmatory model. Whether the text was generated or drawn from the classic set enters as a covariate, and the substantive comparison between the two is a contingent hypothesis, for reasons given under Table 2.

Three variables describe the reader and the reading context. The age band is the platform's own record of the child's age group, reduced to the two bands this study covers. The reading level is the level the family has set for the session, which determines which version of a text is served and

is therefore an attribute of the reading as well as of the reader. Narration is the session-level flag recording that a spoken rendition accompanied the reading. All three are chosen by the family and none is randomised.

Accumulated in-app reading, which serves Hypothesis 6, is the cumulative count of words in completed distinct stories the account read before the current session, on a log scale. Session order and the number of days since the account's previous eligible session enter alongside it, so that accumulated reading is separated from the simple passage of time and from the position of a session in a sequence. Within tier 2 the measure is decomposed into the account's own mean and the current session's deviation from it, and the deviation term is the registered test.

The data are crossed, not nested: readings sit within accounts, a text is met by many accounts and exists in several languages and at several reading levels, and an account may read in more than one language. We will therefore treat Account, Text, Language, Country and App version as crossed grouping factors wherever replication permits, and report the counts supporting each before fitting any model. Reading level, with only three levels, enters as a fixed factor. If at least 90 per cent of texts are read in only one eligible session, a Text varying intercept is not estimable and is replaced by varying intercepts for the repeated attribute combination that generated the text. That rule and any consequence of it will be reported.

## 3.6 Measurement, availability and missingness audit

Before fitting any outcome model, we will run and archive an audit of the frozen extract and its dictionary. It establishes field availability and units, the granularity of the age field, how often a text and a reading level recur, the narration flag's behaviour within accounts and within sessions, duplicate and crash flags, changes across app versions, per-language feature coverage, and every

nesting and crossing count the models require. It also reports missingness for every planned variable, overall and by language, age band, reading level, narration state and session order.

Three rules govern what the audit may change. A feature that is structurally unavailable in a language is not imputed: the language simply does not contribute it. A session missing an outcome is excluded from the model for that outcome alone, and a missing narration flag does not remove a session from a model that has no narration term. No variable is selected, transformed or discarded because of its association with reading time or with the rating.

The rating in Hypothesis 5 is optional and was absent for 25 of the 35 sessions in the structural sample, so high missingness is expected. Where the rating is missing for more than five per cent of otherwise eligible sessions, we will model availability from observed pre-response variables and report an inverse-probability-weighted analysis as a sensitivity check, with weight distributions and common support shown.

## 3.7 Analytic plan

We will fit all confirmatory models as Bayesian hierarchical models with the brms package and its Stan backend[58], standardising continuous predictors within language after transformation. Wherever the accompanying systematic review supports it, weakly informative priors are replaced by priors derived from that review, as described under Priors.

### 3.7.1 Outcome-blind dimension reduction

The observed text features are correlated by construction, because the same decision about what reading level a text is written for moves lexical, grammatical and discourse properties together. Principal component analysis is used to reduce them without treating them as effects of an unobserved

latent cause[22]. It is conducted separately within the lexical, grammatical and discourse families. The affective measures are kept as separate secondary predictors, because valence and arousal are theoretically distinct and need not move together. Narrativity, though computed as a discourse feature, is likewise excluded from the component solution, because it enters every model as a separate secondary predictor and must not sit on both sides. Each distinct text contributes one row regardless of how often it was read, so a popular text does not determine the solution.

Counts are transformed with $\log(1 + x)$ and bounded proportions with an empirical-logit transform where these reduce skew. A feature is dropped from a language if more than 20 per cent of values are missing, if fewer than 80 per cent of texts receive a valid token-level match, or if its standard deviation after transformation is below 0.01. A language contributes to a family's solution only if it has at least 100 distinct eligible texts and at least three valid features in that family, two of them anchors. Where two features correlate at $|r| > .90$, the retained one is chosen in this order: lower missingness, then broader language coverage, then higher extraction reliability on a manually checked sample, then the measure listed first under Materials. No outcome correlation enters any of these decisions.

Features are standardised within language. To stop the largest language determining the solution, a correlation matrix is computed within each eligible language, the correlations are transformed to Fisher's *z*, the matrices are averaged with equal weight per language, and the average is transformed back before the analysis[59]. Sampling adequacy requires individual Kaiser-Meyer-Olkin values above .50 and an overall value above .60. The number of components is set by parallel analysis against the 95th percentile of 10,000 random datasets[60]. The scree plot is reported but does not override that rule. Where a family retains more than one component, promax rotation with $\kappa = 4$ is

used, because linguistic properties may correlate[61]. Regression scoring coefficients from the frozen, language-balanced solution are applied to each text's within-language standardised features.

Each confirmatory component must load at least .30 in absolute value on at least two of its anchors. The anchors are frequency, surprisal and word length for LexicalDifficulty; dependency distance, subordination, clause density, sentence length and noun-phrase depth for GrammaticalLoad; and adjacent-sentence similarity, referential overlap and the density of causal and contrastive connectives for ReferentialContinuity. Age of acquisition contributes to the lexical family where a broad native norm set exists but is not an anchor, because the cross-linguistic set holds only 299 words per language. Connective density enters by relation type, and only the causal and contrastive types are anchors. The restriction is substantive: in 794 readers across 20 texts, causal and contrastive connectives raised comprehension, temporal ones did nothing and additive ones lowered it, so a single mixed count has no defined expected sign[62]. Signs are fixed so that higher LexicalDifficulty and GrammaticalLoad mean a more demanding text and higher ReferentialContinuity a more explicitly connected one. If no component in a family satisfies the rule, the hypotheses resting on that family are reported as untestable. The complete pattern matrix, the component correlations, the variance explained, the language-specific matrices and the loading congruence for the five largest languages are reported. Language-specific and sparse solutions are sensitivity analyses and not alternative routes chosen after seeing an outcome.

As recorded under Data provenance, none of this has been run. The first fitting will follow in-principle acceptance.

### 3.7.2 Confirmatory models

In tier 1, we will estimate a model of within-language standardised log reading time per word with the three components, the age band, the narration flag, the interactions of each component with the age band and with narration, the reading level, log text length, narrativity, valence, arousal and the generated-or-classic indicator as fixed effects. Its grouping structure comprises varying intercepts for Text, Country and App version, varying intercepts and uncorrelated component slopes for Language, and a varying intercept for Account where the estimability rule above is met. Hypotheses 1 to 4 are evaluated in this model.

The Hypothesis 5 model has the same fixed and grouping structure and a cumulative-logit likelihood over the three rating categories, with category thresholds allowed to vary by language so that a difference in how families in different countries use the middle category is not read as a difference in text difficulty.

For tier 2, we will refit the same specification with a varying intercept for Account, decomposing accumulated reading into the account's mean and the current session's deviation from it, alongside session order and days since the previous session. The deviation term is the registered test for Hypothesis 6.

The tier-1 model also carries a distributional sub-model in which the residual standard deviation is itself predicted by the narration flag. The ratio of the two residual standard deviations is the compression diagnostic Hypothesis 4 depends on, and we will report it with that hypothesis.

Posterior predictive checks will assess calibration and residual structure by language, age band, reading level and narration state in every model.

### 3.7.3 Priors

For each confirmatory main effect, priors are derived from the accompanying systematic review through a deterministic pipeline whose rules were frozen before any extraction row existed, and which will be run, with its output archived, before any outcome datum is accessed. The review's extracted estimates for the corresponding predictor family and outcome class are sign-harmonised so that facilitation always means faster reading, converted to a common standardised scale, and pooled by random-effects meta-analysis with one estimate per study per relation. The prior is Student's $t(3, \hat{\mu}, \sigma)$, where $\hat{\mu}$ is the pooled estimate translated to the model's scale and $\sigma$ is twice the combined pooling and between-study uncertainty, with a floor of 0.10. The doubling and the floor guard against the gap between a laboratory comprehension test and a passively timed reading in a game.

Coefficients without adequate review evidence, which means every interaction term and any family supported by fewer than two studies, keep weakly informative priors: $t(3, 0, 0.5)$ for standardised main effects and $t(3, 0, 0.25)$ for interactions. Group-level and residual standard deviations take half-$t(3, 0, 1)$ priors and any estimated correlation matrix takes LKJ($\eta$ = 2). Intercept priors are calibrated by prior-predictive simulation to the outcome scale, without reference to any predictor-outcome association. The derivation rules are summarised in Supplementary Section S6, and the full prior table with its forest plots will be archived before the extract is transferred and will accompany the Stage 2 report. We will refit every model with weakly informative priors throughout, and with the derived scales halved and doubled, and report conclusions under each.

### 3.7.4 Decision rules and diagnostics

For every confirmatory coefficient, we will report the posterior median, the 95 per cent highest-density interval, the probability of the predicted direction, the proportion of the posterior inside

the region of practical equivalence, and a model-implied contrast on an interpretable scale. The region is $|\beta| < 0.05$ standard deviations for a one-standard-deviation change in a predictor of reading time, and an odds ratio between 0.95 and 1.05 for the rating model. More than 95 per cent of the posterior inside the region is read as practical equivalence, less than 5 per cent inside it as a practically non-negligible association, and everything else as inconclusive.

We will run four chains of 4,000 iterations, half of them warm-up. Acceptable sampling requires $\hat{R}$ below 1.01, bulk and tail effective sample sizes above 1,000 for every confirmatory parameter, no divergent transitions, satisfactory energy diagnostics and no unresolved maximum-treedepth warnings. Models failing these will be refitted with a higher target acceptance rate, greater tree depth and more iterations. If simplification is still needed, language-level slopes are removed in the fixed order lexical, then discourse, then grammatical, and every estimable varying intercept is retained. Every simplification and its diagnostic consequence will be reported. Variance inflation factors will be reported for the fixed-effect design matrix. A high value does not trigger an outcome-dependent deletion: the joint model remains primary, and single-family models are reported as a collinearity sensitivity analysis.

For Hypotheses 2, 3 and 4, we will compare the model with one omitting the relevant interaction, using Pareto-smoothed importance-sampling leave-one-out cross-validation, and report the expected log pointwise predictive density difference, its standard error and the Pareto-$k$ diagnostics, with exact refits where importance sampling is unreliable. Predictive comparison supplements the coefficient-level summaries and does not gate them.

### 3.7.5 Outcome-neutral quality checks

Six checks run before the confirmatory models are interpreted, each with a stated pass criterion and a stated consequence of failure. The first four and the sixth are decided without reference to any confirmatory coefficient. The fifth is itself a registered clause of Hypothesis 5 and is evaluated with it.

The first is a permutation null. We will reassign texts at random to readings within language and reading level, and refit the tier-1 model 200 times. Every text coefficient must be practically equivalent to zero in at least 95 per cent of those refits. A coefficient that survives permutation means the pipeline is carrying structure that does not come from the pairing of a text with a reading, and we will withhold the confirmatory results until the leak is found and reported.

The second is a positive control on the timing column. Total elapsed time must rise with the story's word count, with more than 99 per cent of the posterior above zero. This is close to a tautology if the column means what the dictionary says it means, which is why its failure would be informative: a timing field dominated by idle or backgrounded time need not track how much text there was to read.

The third asks whether the association is an artefact of platform assignment. We will re-estimate the three text coefficients within each reading level separately, and each must retain at least half the magnitude it had in the pooled model. The account on which text difficulty is merely assigned to children who differ in capacity predicts that these coefficients shrink towards zero when the level is held fixed. The account this protocol advances predicts that they do not.

The fourth is a set of three independence claims the theory object entails: conditional on the reading

level and on whether a text was generated or classic, the three components should be close to independent across texts, narration should be unrelated to accumulated reading, and the difficulty rating should carry no further information about reading level once reading time is conditioned on. We will report each as an estimate with an interval. A clear failure does not invalidate the confirmatory tests, but it identifies a path the causal diagram omits, and it is reported as such.

The fifth is convergent validity, the ordering clause of Hypothesis 5, specified there and evaluated with it.

The sixth is a validation of the feature pipeline itself. We will check a stratified random sample of 50 texts per language by hand for tokenisation, sentence splitting and dependency attachment, and report the agreement statistics per language before any feature is joined to an outcome. A language whose labelled-attachment agreement with the hand-checked parse falls below .85 contributes to descriptive reporting but not to the grammatical component.

### 3.7.6 Design and precision analysis

Because we will analyse every eligible record, the question is not how large a sample to collect but what the available records can settle. Here that turns on a single quantity: the text coefficients are identified between distinct texts, and how many distinct texts the extract holds is unresolved.

We therefore simulated the planned crossed structure from a portable specification, sweeping the number of distinct texts at constant readings per text, and separately sweeping readings per text at a constant number of texts[63]. Simulation was on the within-language standardised scale, with the text intercept contributing 9 per cent of the variance, the child intercept 20 per cent and the residual the remainder, so a simulated coefficient is directly comparable with the region of practical equivalence. The simulation generated child variance and then handed the analysis a dataset in

which each account contributes one reading, so that variance reached the fitted model as inflated residual. That is the less favourable of the two cases the estimability rule allows, so the requirements below are an upper bound on what the design needs. A coefficient the size of the region can never be declared determinately outside it, so we ran the sweep at twice that size, the smallest effect this protocol treats as worth reporting. A null condition with every coefficient set to zero established how reliably the design can declare practical equivalence, so an inconclusive verdict is predictable in advance.

For the three text coefficients, an effect of 0.10 standard deviations reached a determinate verdict nine times in ten at 738 distinct texts for lexical difficulty (95 per cent interval 640 to 843), 660 for grammatical load (593 to 731) and 634 for referential continuity (568 to 704). The interactions with the age band needed roughly twice as many: 1,230 texts for grammatical load (1,107 to 1,359) and 1,431 for referential continuity (1,276 to 1,596). The narration contrast needed 1,234 texts (1,113 to 1,361) before the design effect for imbalance is applied. Declaring practical equivalence when there is nothing there cost about the same: at 800 texts a main effect was correctly declared equivalent in 87 to 90 per cent of replicates and at 1,600 in every one, while the interactions reached 89 to 92 per cent only at 1,600. An inconclusive verdict is therefore predictable from the number of texts alone. The within-child term needed 563 children in tier 2 (463 to 674) contributing five readings each.

The second sweep is the more instructive. Holding the number of distinct texts at 250 and raising readings per text from 2 to 60 narrowed the interval on a main effect from 0.19 to 0.08 standard deviations, yet never carried it to a determinate verdict, reaching 0.70 at best. Over the same range the age-band interactions climbed from under 0.11 to between 0.87 and 0.90. Sessions and texts are therefore not interchangeable: a text coefficient is bounded by how many different texts the

platform generated, whereas an interaction between a text property and a reader attribute gains from many different children meeting the same text. This matters because sessions are the quantity this platform has in abundance and distinct texts are the quantity in doubt, so a very large export could still leave Hypothesis 1 underpowered while answering Hypotheses 2 and 3.

Figure 3 shows both sweeps. The threshold each hypothesis has to clear is recorded in Table 1 and was fixed before the extract was released. A hypothesis whose required contrast is absent, or whose simulated interval cannot separate the region's boundary from the effect, will be labelled untestable, or expected to be inconclusive, before its model is fitted.

Narration requires one further step. The simulation balances the narration contrast, whereas narration was on in 32 of the 35 sessions of the structural sample. An unbalanced binary contrast at prevalence $p$ inflates the variance of its coefficient by $1/(4p(1-p))$ relative to balance, so we will multiply the simulated requirement by that factor at the prevalence the provider-computed counts report, and state the adjusted requirement before fitting Hypothesis 4. At a prevalence of 0.09 the factor is about 3.0.

### 3.7.7 Sensitivity analyses

The following sensitivity analyses are pre-specified. We will refit the timing models with no plausibility bounds at all, with winsorisation at the 1st and 99th and at the 0.5th and 99.5th percentiles within language, and with posterior-predictive residual trimming. Where the censoring rule is triggered, we will report the censored and uncensored fits side by side. We will vary the tier-2 threshold from three distinct texts to two and to five, and add repeated readings with a repetition indicator. We will restrict analyses to the five largest languages, and fit language-specific models and component solutions with loading congruence reported. We will replace the component scores with

sparse-component scores and with pre-specified equal-weight composites of the anchor features. For narration, we will restrict the comparison to accounts observed in both states, and adjust narrated timing for audio duration if that duration is ever supplied. We will compare the rating model under adjacent-category and cumulative-logit links, stratify it by language, and compare the inverse-probability analysis for rating non-response with the complete-case estimates. We will compare Text grouping with grouping on the attribute combination that generated the text. Finally, we will halve and double the prior scales and apply the alternative regions of practical equivalence, 0.03 and 0.10 standard deviations for reading time and odds-ratio bounds of 0.97 to 1.03 and 0.90 to 1.10 for the rating.

To ease comparison with the prior literature, we will report maximum-likelihood mixed models with the same fixed and estimable grouping structure in a supplementary table. They are descriptive, carry no confirmatory weight, and will not be used to resolve a Bayesian result.

## 3.8 What is not yet settled, and what follows from each answer

We put thirteen questions about the extract to the data controller on 7 August 2026, and they remain open at submission. Leaving them open is not a gap in the protocol so long as each has a decision rule attached in advance, and Table 2 attaches one to each. Every rule below is a function of the answer alone.

Five of these questions correspond to auxiliary assumptions recorded in the theory object, which is how the protocol keeps the statistical contingency and the theoretical commitment in the same place. If the answer to a question falsifies its assumption, the predictions that assumption protects are the ones the table reassigns.

**Table 2**

*Questions outstanding at submission and the decision rule attached to each*

| What is unresolved | What it bears on | What the protocol does under each answer |
|---|---|---|
| How many distinct texts the extract holds, settled by the provider-computed counts together with question d. The controller's own estimate spans a few hundred to a few million, according to whether one combination of chosen attributes yields one shared text or a fresh text per session. | Every text coefficient, since all are identified between texts. | The count is read from the provider-computed figures before modelling and located on the precision curve in Figure 3. Below the threshold in Table 1 a hypothesis is declared underpowered before its model is fitted. Where each session carries its own text, the Text varying intercept is replaced by varying intercepts for the repeated attribute combination, under the sparsity rule given under Measures, and precision is then governed by the number of attribute combinations. We will report which case obtains. |
| Whether the narration flag records that audio played or only that it was available, and whether it can change within a session (question a). | Hypothesis 4, through the auxiliary assumption that narration played. | If it records playback, Hypothesis 4 stands as written. If it records availability only, the variable is relabelled as availability, the contrast becomes an intention-to-treat comparison whose attenuation is diluted towards the null by non-compliance, and that dilution is stated wherever the coefficient is reported. Sessions in which the flag changes are excluded and counted. |
| Whether one story identifier maps to one text, and whether a text differs across reading languages by translation or by independent generation (question d). | The identity of the stimulus, through the auxiliary assumption that the keyed text is the text the child read, and therefore every model's grouping structure. | If one identifier is one text, texts are counted by identifier crossed with language and reading level, as specified. If a text is translated across languages, a varying intercept for the underlying story is added, crossed with Language. If each language is generated independently, texts in different languages are distinct and no such term is added. |
| What the reading-time interval covers: whether it includes embedded activities, and whether it includes time backgrounded or idle (question c). | The outcome itself, through the auxiliary assumption that elapsed time indexes effort. | If the interval excludes them, the outcome stands as written. If it includes embedded activities, texts that offer them are identified from the story attributes, an indicator enters every model, and the primary analysis is repeated within texts that offer none. If it includes idle or backgrounded time, the upper plausibility bound and the censoring rule carry more of the work, and the limitation is stated in the abstract itself. |
| Whether an account persists across reinstalls and whether siblings can share one (question f). | Tier 2 and Hypothesis 6, through the auxiliary assumption that an account is a child. | If an account can be shared, the within-child reading of Hypothesis 6 is withdrawn in advance and the coefficient is reported as a within-account association, with the ambiguity stated in the hypothesis itself. |

| What is unresolved | What it bears on | What the protocol does under each answer |
|---|---|---|
| Whether the reading level can change within an account over time, and whether it changes the text served or only the page count (question g). | The platform-assignment confound and the third quality check. | If it changes only pagination, texts are comparable across levels and the pooled estimate is primary with the within-level estimate as a check. If it changes the text, the within-level estimate becomes primary and the pooled one is reported alongside it. Where the level changes within an account, that change is used as a within-account contrast in an exploratory analysis only. |
| The source of the age band and how much of it is missing across the whole window (question h). It was blank for 10 of 35 sessions in the structural sample. | Hypotheses 2 and 3. | Missingness is reported by language, reading level and app version. Above 20 per cent, availability is modelled from observed pre-response variables and an inverse-probability-weighted fit is reported alongside the complete-case fit, with both interpreted together. |
| Which reading languages the window covers. The app is marketed in 19 languages and the structural sample showed ten (question i). | Coverage of the feature pipeline and the language grouping factor. | The observed set is reported as found. A language for which the required resources do not exist contributes to the participant-flow diagram and the descriptive tables but not to any feature-dependent model, and the excluded volume is quantified. |
| Whether the classic stories are fixed, human-authored texts identical for all users within a language (question j). | The comparison between machine-generated and human-authored material, contingent hypothesis C4. | Only if they are fixed and identical does C4 become confirmatory, on the activation rule below. Otherwise the generated-or-classic indicator remains a covariate and the comparison is reported as exploratory. |
| Whether unfinished sessions are retained anywhere, and what the completion rate is (question k). | Generalisability, since every row in the extract is a completed reading. | The study describes completed readings and says so. Where an aggregate count of unfinished sessions by language is available, the completion rate is reported and the restriction quantified. |
| What fields the older data carry and what window they cover (question l). | Hypothesis 6, through the within-window assumption that accumulated reading acts on decoding automaticity over the export window and not the reverse. The measure is truncated by a window opening on 1 January 2026. | If an older file carries the same fields, accumulated reading is computed over the longer window and the analysis is repeated on both. Otherwise the measure is left-truncated at the window's opening and is described throughout as reading accumulated within the observation window, which makes an experienced reader look new. |
| Whether post-story responses include comprehension items or only the difficulty rating (question m). | Contingent hypotheses C1 to C3. | If item-level comprehension responses exist, C1 to C3 activate exactly as written below, on the activation rule and deadline stated there. If they do not, they are reported as not activated, and no comprehension claim is made anywhere in the paper. |

| What is unresolved | What it bears on | What the protocol does under each answer |
|---|---|---|
| What the per-session activity counters measure. In 6 of the 14 structural-sample sessions with activity, they sum to more than the whole reading duration, in one case 438 seconds against 222 (question b). | Nothing, by decision. | These fields are set aside and are not used as a predictor, a covariate or an exclusion criterion. The decision is recorded here so that their later use would be visible as a deviation. |

## 3.9 Contingent hypotheses

We specify four further hypotheses in full here, outside the confirmatory set, because each depends on a fact about the extract that the data controller has not yet confirmed. Each carries an activation rule, and the rule is a function of that fact alone.

A contingent hypothesis activates only if its condition is confirmed in writing by the data controller before this protocol is submitted, and the confirmation is reproduced in the supplementary material. A condition confirmed after submission does not activate anything: the hypothesis is then reported at Stage 2 as exploratory, under that label, and is not counted among the confirmatory tests. This asymmetry is deliberate, because a hypothesis activated after review has not been reviewed.

**C1. Coherence and inferential comprehension.** Condition: item-level comprehension responses exist and carry a recoverable question type. We would predict that, in an item-level model of response correctness, ReferentialContinuity × QuestionType is positive with inference questions coded 1 and literal questions 0, so that surface continuity helps more where the question requires integration[28,29,64].

**C2. Narration and comprehension.** Condition: item-level comprehension responses exist, and accounts are observed both with and without narration. We would predict that, within an account, narration is positively associated with response correctness after text, age band, reading level, accumulated reading and language are adjusted for. The expected size is small[34,35], and the coefficient identifies an association and not an effect.

**C3. Reading pace and comprehension.** Condition: as C2, restricted to sessions without narration. We would predict that an account's mean log reading time per word is negatively associated with

response correctness, so that accounts with a faster habitual pace answer more accurately[40]. The simultaneous within-account deviation term keeps this between-account association distinct from a difficult or interrupted session.

**C4. Machine-generated against human-authored text.** Condition: the classic stories are confirmed to be fixed, human-authored texts, identical for all users within a language. We would predict that the three text components predict reading pace with coefficients of the same sign and of comparable magnitude in generated and in classic texts, and that the interaction of each component with the generated-or-classic indicator is practically equivalent to zero. This is a test of exchangeability, not of quality. A non-negligible interaction would mean that the properties governing how children read human-written stories govern their reading of generated ones differently, which matters well beyond this platform.

## 3.10 Randomisation, blinding and analytic independence

Nothing in this study is randomised. It is a secondary analysis of records generated by families using a commercial application, and no allocation, assignment or condition is under anyone's control. The only random assignment anywhere in the protocol is the permutation quality check, where we reassign texts to readings at random precisely to destroy the association under test.

Blinding is structural: we cannot see the outcome data because the data controller holds them, under the sequestration arrangement described above. We will write, run and debug the outcome-model code against data simulated from the design specification deposited with this protocol, so that every script is complete and exercised before the real file exists on any machine of ours.

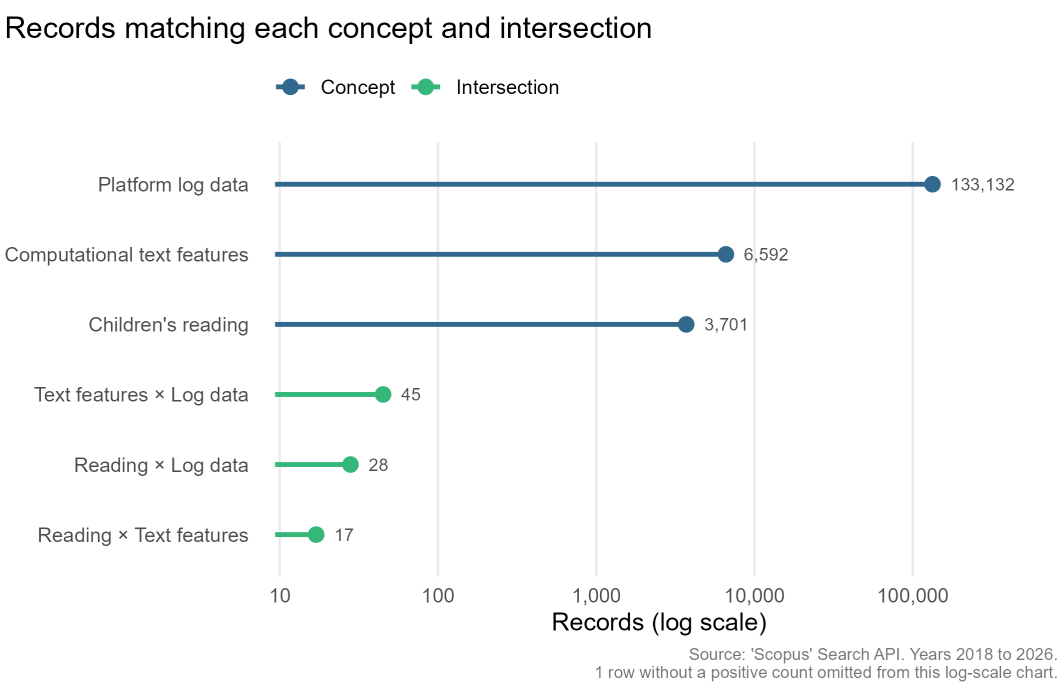



Figure 1: Sizing the literature this study sits in. Records on Scopus for 2018 to 2026, for children's reading comprehension, computational text features and platform log data, and for every intersection of the three. The pairwise intersections hold 17, 28 and 45 records. The three-way intersection, which is where this study sits, holds none, and so does not appear on the logarithmic axis. Search expressions and the date of the search are in the supplementary record.

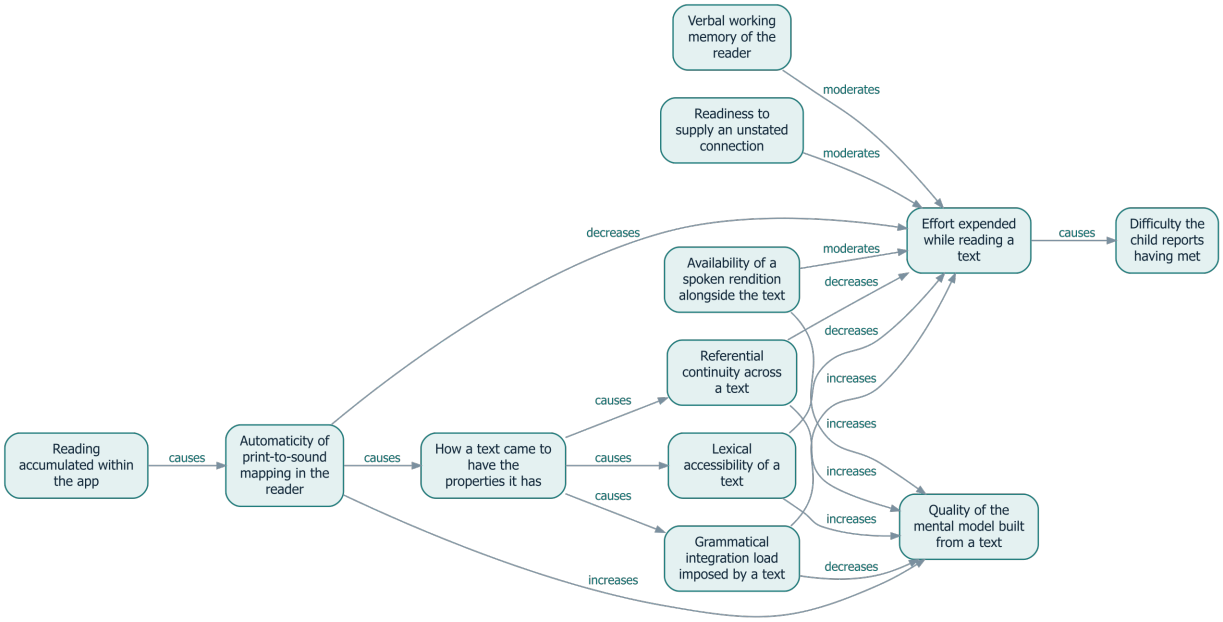



Figure 2: The nomological net, generated from the deposited theory object. Text properties and reader capacities act on the effort a reading costs, which in turn shapes the difficulty the child reports. A parallel set of paths reaches the quality of the mental model built, which these data can measure only if comprehension items prove to exist. Text provenance, meaning the generation route and the reading level being served, is a common cause of all three text properties and receives a path from the reader's own decoding automaticity, which is why every text coefficient is estimated within reading level.

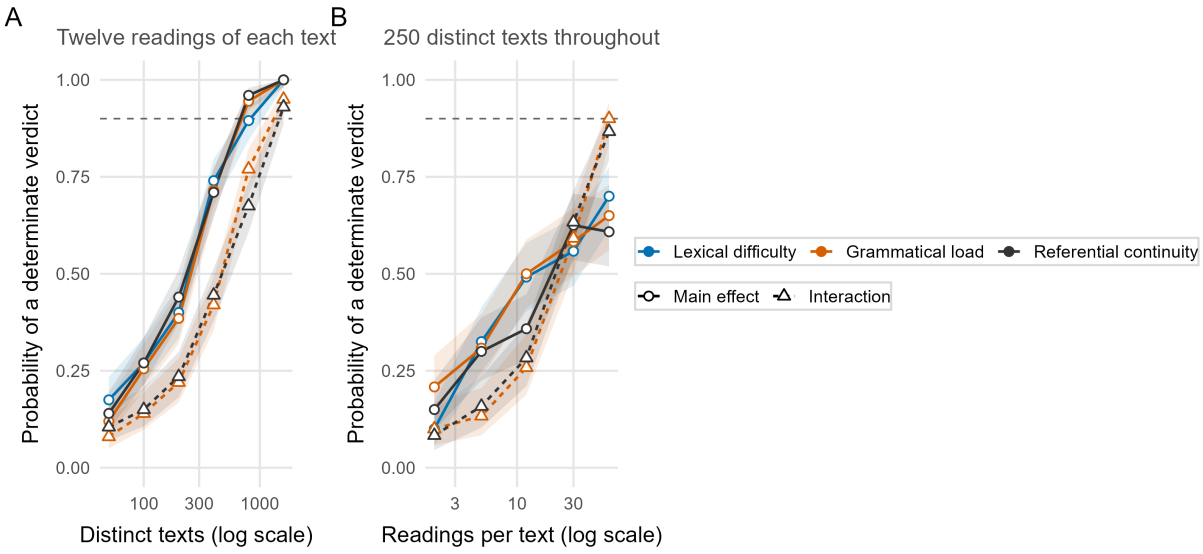



Figure 3: Precision as a function of what there is more of. The left panel sweeps the number of distinct texts at twelve readings each, and the right panel holds the number of texts at 250 and sweeps readings per text. Points are the probability that a 95 per cent interval falls determinately outside the region of practical equivalence for an effect of 0.10 standard deviations, over 200 replicates, with Wilson intervals. The two panels answer different questions, and only one of them is about having more data of the kind this platform generates most readily.

## 3.11 Figures

## 3.12 Data and code availability

The session-level extract is held by the data controller and is not in our possession at submission. On in-principle acceptance it will be transferred under a data-sharing agreement between the University of Oxford and the data controller, whose terms govern what may subsequently be published. The story texts, the computed text features, the component scores, the fitted model objects and every table and figure underlying the reported results will be deposited on acceptance, to the extent the agreement permits. Where a field cannot be released, we will state the reason and deposit a derived form sufficient to reproduce the reported analyses. The provider-computed aggregate counts and the file manifest are deposited with this protocol, since they contain no record of any individual.

All analysis code is public and version-controlled. The theory object with its checks and derived diagram, the design specifications with the precision analysis, and the literature sizing are deposited with this protocol, so everything except the fitting of the models to real data can be reproduced now. The outcome-model code will be written against simulated data, as described above, and deposited before the extract is transferred. The four packages this protocol relies on are open source and archived: theoryforge for the theory object and its rigour scoring, pilotr for the design specification and the precision analysis, scopusflow for the literature sizing, and depictr for the figures[41,63,65,66]. Each is cited by its concept identifier, and the archived releases hold the versions used here, which are also recorded in the session records deposited beside every output.

## 3.13 Reporting summary

Further information on research design is available in the Nature Portfolio Reporting Summary linked to this article.

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